EXTENDING GALILEO

THE PROBLEM

Most Arduino boards use an 8-bit AVR architecture based microcontroller manufactured by Atmel. The fastest microcontroller used on Arduino runs at 16MHz. The only way of communicating with the microcontroller is by writing a sub-program to communicate over the serial interface. Such a sub-program will itself interfere with the execution of the main program, thereby reducing the speed of execution of the main task.

Thus, these boards are suitable for performing small tasks and there arises the need to interface multiple boards together, distributing small chunks of load among individual boards.

THE SOLUTION

With increasing possibilities and hardware capabilities, makers now need hardware with greater computing power that can handle tasks efficiently reducing hardware complexity and footprint. Computer processor manufacturer, Intel comes to the rescue here with hardware by the name of Intel Galileo – a good step towards giving hardware hackers the capability to build the Internet of Things.

INTEL GALILEO SPECIFICATIONS

Named after the "Father of Modern Science", Galileo Galilei, Intel Galileo is a microcontroller board based on the Intel® Quark SoC X1000 Application Processor, a 32-bit Intel Pentium based x86 system on a chip running at 400MHz with 512KB SRAM that supports Complex Instruction Set Computing (CISC). It is the first Arduino compatible board built on Intel® x86 architecture designed for hardware and software pin-compatibility with various shields designed for the Uno R3 version of Arduino. The Galileo board uses the Arduino Software Development Environment for application development. Check these links for more info on the boards: http://dgit.in/galqark, http://dgit.in/galfull

In addition to compatibility with the Arduino, the Galileo board has several standard I/O ports used in the computer industry and features to expand its capabilities beyond the Arduino ecosystem such as:

- mini-PCI Express slot
- 100Mb Ethernet port
- Micro-SD slot
- RS-232 Serial port
- USB Host port
- USB Client port
- 8 MByte NOR flash on the board

> The maker culture is a culture representing a technology counterpart of DIY culture. Makers are aligned to engineering-oriented activities such as electronics, robotics, 3-D printing and metal/woodworking. The culture stresses on innovation of newer applications for betterment of society using available technologies. Using available skills practically to build systems is what makers do the best.

The Intel processor and I/O capabilities of the chip provide a highly capable product for students as well as the maker community. It will therefore be useful to advanced developers seeking a simple and cost effective development environment over the more complex Intel® Atom™ or Intel® Core™ architecture based designs.

The Galileo board is a freely available open source hardware design. Design schematics and other details are available for download, thus facilitating easy hacking if need be. A list of shields supported by Galileo and other documentation is available at http://dgit.in/intelcommunities

WHY GALILEO?

Intel Galileo combines the computing power of the Intel Quark processor with the simplicity of the Arduino platform. The Arduino platform enables anyone with the knowledge of C/C++ to program hardware. The Galileo platform combines the ease of Arduino's hardware capabilities with the power of a fully operational Linux operating system. Many sketches (Arduino programs) written for Arduino boards can be ported over to the Galileo with little or no modification. All popular Arduino libraries dedicated to SD card, Ethernet, Wi-Fi, EEPROM, SPI and I2C/TWI devices are available along with access to the Linux OS on the board using system() function calls.

INSTALLING GALILEO

SOFTWARE

The software used to program the Galileo is the custom-tailored Galileo version of Arduino Integrated Development Environment (IDE) and can be downloaded at http://dgit.in/intelcommunitiesdoc22226 for your operating system platform viz. Windows, Mac OS X and Linux.

ARDUINO INSTALLATION

Windows:

Unzip the downloaded file and open the file 'arduino.exe' in the extracted folder.

Mac OS:

Unzip application and move it into your Applications folder. Open the 'Arduino' application.

Linux:

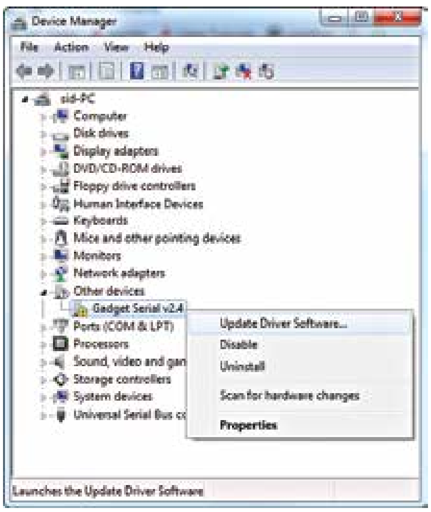
GUI: Extract the folder and open the file 'arduino' (making it executable from Properties, if necessary).

Command line: Change directory 'cd' to the download directory. Use the command 'tar -zxvf arduino-1.5.3-linux32.tar.gz' to extract the directory. Use 'cd' to open the extracted folder and './arduino' to open the required file.

DRIVER INSTALLATION

Windows:

1. Connect the board to the computer.
2. Open Device Manager (Open 'Control Panel' > Select 'System' > Click 'Device Manager'.)
3. Right-click 'Gadget Serial v2.4' device under the 'Other devices' tree > Select 'Update Driver Software'
4. On the window that pops up, click 'Browse my computer for driver software' > Select 'Browse...' > Navigate to 'hardware\arduino\x86\tools' folder within your Arduino Galileo software installation > Click 'Next'.
5. Click 'Install' on the next Windows Security window that pops up.
6. The Device Manager will get updated. Look under the 'Ports' tree now. There should be an entry for 'Galileo (COM #)'
7. Driver installation for Intel Galileo is now complete.



Device Manager before Galileo Driver Installation